Chapter 14: Heat and Heat Transfer Methods

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PHY201

Lecture 14

Part I

Heat and Specific Heat

Outline of Part I: Heat and Specific Heat

What is Heat?

Specific Heat Capacity

Heat

Heat (Q) — SI Unit: joule (J)

Heat is **energy transferred between objects due solely to a temperature difference.**

Heat flows spontaneously from higher temperature to lower temperature.

Heat is NOT

- The temperature of an object.
- The internal energy stored in an object.
- A substance or fluid inside an object.

Heat IS

- Energy *in transit* between objects.
- Only defined during transfer — an object does not "contain" heat.
- Measured in joules (J).

Units of Heat

Joule (SI unit)

The SI unit of heat is the joule (J) — the same unit as any form of energy.

Calorie

- 1 calorie (cal) = heat required to raise 1 g of water by 1 °C.
- 1000 = 1 k = 1 Calorie (Cal, dietary)
- 1 k = 4186 J

The Calorie Confusion

- Nutritionists use the **Calorie** (capital C) = 1 kcal = 4186 J.
- Food labels say "280 Calories" but mean 280 k.
- $280\,\text{k} = 280 \times 4186\,\text{J} = 1.17 \times 10^6\,\text{J} = 1.17\,\text{MJ}$ — enough to lift a 1000 kg car by 119 m.

Check Your Understanding

A bag of chips contains 150 Calories (dietary).

(a) How many joules of energy is this?

(b) If all this energy went into lifting a 70 kg person, how high would they be lifted?

Come to lecture to see the answer.

Outline of Part I: Heat and Specific Heat

What is Heat?

Specific Heat Capacity

Temperature Change and Heat Capacity

$Q = mc\Delta T$ — Heat Equation (No Phase Change)

$$Q = mc\Delta T$$

- Q: heat added to (or removed from) sample (J)
- m: mass of sample (kg)
- c: specific heat of substance (J/kg·°C)
- $\Delta T = T_f - T_i$: temperature change (°C or K)

- $Q > 0$: heat flows *into* the sample (T increases).
- $Q < 0$: heat flows *out* of the sample (T decreases).
- Valid only while the substance remains in a **single phase** (no melting or boiling).

Specific Heat Capacity

Specific Heat c — SI Unit: $\frac{\mathrm{J}}{\mathrm{kg}\,^\circ\mathrm{C}}$

Specific heat is a measure of a substance's **resistance to temperature change**.

A substance with high c requires more heat to change its temperature by $1\,^\circ\mathrm{C}$.

Water has an exceptionally high specific heat ($c = 4186\,\frac{\mathrm{J}}{\mathrm{kg}\,^\circ\mathrm{C}}$), which makes it ideal as a coolant (car radiators, body temperature regulation) and explains why coastal climates are milder than inland climates.

Specific Heat Values

Substance	c **(J/kg·°C)**	c **(kcal/kg·°C)**
Water (liquid)	4186	1.000
Steam	2010	0.480
Ice	2090	0.500
Aluminum	900	0.215
Glass	840	0.200
Iron/Steel	452	0.108
Copper	385	0.092
Silver	235	0.056
Lead	128	0.031

Note: $1\,\frac{\text{J}}{\text{kg}\,{}^\circ\text{C}} = 1\,\frac{\text{J}}{\text{kg}\,\text{K}}$ since a change of $1\,{}^\circ\text{C}$ equals a change of $1\,\text{K}$.

Example: Heating Water

How much heat must be supplied to a 200 mL sample of water to raise its temperature from 13 °C to 27 °C?
$(c_w = 4186\,\frac{\text{J}}{\text{kg}\,^\circ\text{C}},\ \rho_w = 1000\,\frac{\text{kg}}{\text{m}^3} = 1\,\frac{\text{g}}{\text{mL}})$

$$m = \rho V = 1\,\frac{\text{g}}{\text{mL}} \times 200\,\text{mL} = 200\,\text{g} = 0.200\,\text{kg}$$

$$\Delta T = 27 - 13 = 14\,^\circ\text{C}$$

$$Q = mc\Delta T$$

$$Q = 0.200\,\text{kg} \cdot 4186\,\frac{\text{J}}{\text{kg}\,^\circ\text{C}} \cdot 14\,^\circ\text{C}$$

$$\boxed{Q = 11\,720\,\text{J} \approx 11.7\,\text{kJ}}$$

Since $m = \rho V$:

$$Q = \rho V c \Delta T$$

This alternate form is useful when volume (not mass) is given.

Calorimetry: Heat Exchange Between Objects

Conservation of Energy in a Thermally Isolated System

When two objects at different temperatures reach thermal equilibrium, the total heat exchanged is zero:

$$\Sigma Q = 0 \qquad \Rightarrow \qquad Q_{\text{gained}} + Q_{\text{lost}} = 0$$

i.e. heat lost by the hotter object = heat gained by the cooler object.

- "Gained" and "lost" are encoded in the *sign* of $Q = mc\Delta T$.
- If $T_f < T_{\text{initial}}$ for object A, then $Q_A < 0$ (object A lost heat).
- If $T_f > T_{\text{initial}}$ for object B, then $Q_B > 0$ (object B gained heat).
- The common final temperature T_f is the unknown.

Example: Iron Block Placed in Water

A 10.0 kg iron block at 110 °C is placed in 2.00 L of water at 25.0 °C. Find the final equilibrium temperature T_f.
$\left(c_{\mathrm{Fe}} = 452\,\frac{\mathrm{J}}{\mathrm{kg}\,^\circ\mathrm{C}},\ c_w = 4186\,\frac{\mathrm{J}}{\mathrm{kg}\,^\circ\mathrm{C}}\right)$

$$m_w = 1\,\frac{\mathrm{kg}}{\mathrm{L}} \times 2.00\,\mathrm{L} = 2.00\,\mathrm{kg}$$

$$\Sigma Q = 0: \qquad Q_w + Q_{\mathrm{Fe}} = 0$$

$$m_w c_w (T_f - T_{w,i}) = -m_{\mathrm{Fe}} c_{\mathrm{Fe}} (T_f - T_{\mathrm{Fe},i})$$

$$T_f = \frac{m_w c_w T_{w,i} + m_{\mathrm{Fe}} c_{\mathrm{Fe}} T_{\mathrm{Fe},i}}{m_w c_w + m_{\mathrm{Fe}} c_{\mathrm{Fe}}}$$

$$T_f = \frac{(2.00)(4186)(25.0) + (10.0)(452)(110)}{(2.00)(4186) + (10.0)(452)}$$

$$\boxed{T_f = 54.8\,^\circ\mathrm{C}}$$

Note $T_f = 54.8\,^\circ$C, which is between the two initial temperatures (25 °C and 110 °C), as expected.

The large mass and high c of water means T_f is closer to the water's initial temperature.

Check Your Understanding

You have equal masses of aluminum ($c = 900\,\frac{\text{J}}{\text{kg}\,^\circ\text{C}}$) and copper ($c = 385\,\frac{\text{J}}{\text{kg}\,^\circ\text{C}}$) at the same initial temperature.

If you supply the same amount of heat Q to each, which ends up at a higher temperature?

Come to lecture to see the answer.

Check Your Understanding

A $0.500\,\mathrm{kg}$ copper block ($c = 385\,\frac{\mathrm{J}}{\mathrm{kg}\,^\circ\mathrm{C}}$) at $95.0\,^\circ\mathrm{C}$ is dropped into $1.00\,\mathrm{kg}$ of water at $20.0\,^\circ\mathrm{C}$.

Find the equilibrium temperature T_f.

Come to lecture to see the answer.

Part II

Phase Change and Latent Heat

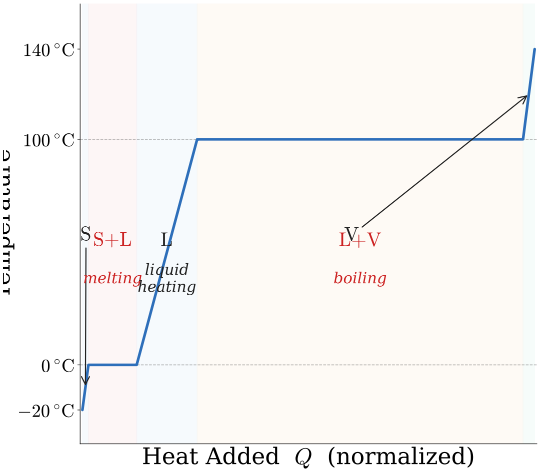
Outline of Part II: Phase Change and Latent Heat

Latent Heat

Outline of Part II: Phase Change and Latent Heat

Latent Heat

Phase Changes Occur at Constant Temperature



Phases in the Diagram

- S: solid (heating ice)
- S+L: melting (flat at 0 °C)
- L: liquid (heating water)
- L+V: boiling (flat at 100 °C)
- V: vapor (heating steam)

Note: $c_{\text{ice}} \neq c_{\text{water}} \neq c_{\text{steam}}$ — different slopes in each region.

When heat is added during a phase change, temperature does *not* increase. Energy goes into **breaking molecular bonds**, not increasing kinetic energy.

$Q = mc\Delta T$ is only valid when *no phase change* occurs. Use a different equation when a phase change occurs.

Latent Heat

$Q = mL$ — Heat During Phase Change

$$Q = mL$$

- Q: heat supplied to cause the phase change (J)
- m: mass of sample (kg)
- L: latent heat coefficient of the substance (J/kg) — tabulated

L_f — Latent Heat of Fusion

- Heat to melt (or freeze) a substance.
- Solid $\leftrightarrow$ Liquid.
- Water: $L_f = 334\,000\ \frac{\mathrm{J}}{\mathrm{kg}}$

L_v — Latent Heat of Vaporization

- Heat to vaporize (or condense) a substance.
- Liquid $\leftrightarrow$ Vapor.
- Water: $L_v = 2.26 \times 10^6\ \frac{\mathrm{J}}{\mathrm{kg}}$

Latent Heat Values

Substance	Melt pt. (°C)	L_f (kJ/kg)	Boil pt. (°C)	L_v (kJ/kg)
Water	0	334	100	2256
Ethanol	−114	104	78	854
Mercury	−38.9	11.4	357	272
Nitrogen	−210	25.5	−196	201
Lead	327	24.5	1750	871

Water's latent heat of vaporization ($L_v = 2256\,\frac{\text{kJ}}{\text{kg}}$) is exceptionally large. Evaporating sweat removes far more energy per kg than simply warming water to body temperature, making sweating an efficient cooling mechanism.

Total Heat with Multiple Stages

Calculating Q_{total} across Temperature Changes and Phase Changes

If a substance passes through both temperature changes and phase changes, sum contributions from each stage:

$$Q_{\text{total}} = \sum_i Q_i = Q_{\Delta T,1} + Q_{\text{LH}} + Q_{\Delta T,2} + \cdots$$

- Use $Q = mc\Delta T$ for each constant-phase segment (slope in T-vs-time graph).
- Use $Q = mL$ for each phase change (flat segment in T-vs-time graph).
- Use c_{ice}, c_{water}, c_{steam} separately — they differ.
- Always work left to right on the T-vs-time graph.

Example: Melting Ice at $-5\,°\mathrm{C}$

How much heat is required to completely melt 1.00 kg of ice that starts at $-5.00\,°\mathrm{C}$?
$(c_{\text{ice}} = 2090\,\frac{\mathrm{J}}{\mathrm{kg}\,°\mathrm{C}},\ L_f = 334\,000\,\frac{\mathrm{J}}{\mathrm{kg}})$

Two stages: (1) heat ice to $0\,°\mathrm{C}$, (2) melt ice.

$$
\begin{aligned}
Q_1 &= mc_{\text{ice}}\Delta T \\
&= (1.00\,\mathrm{kg})(2090\,\tfrac{\mathrm{J}}{\mathrm{kg}\,°\mathrm{C}})(5.00\,°\mathrm{C}) \\
&= 10\,450\,\mathrm{J}
\end{aligned}
$$

$$
\begin{aligned}
Q_2 &= mL_f = (1.00\,\mathrm{kg})(334\,000\,\tfrac{\mathrm{J}}{\mathrm{kg}}) \\
&= 334\,000\,\mathrm{J}
\end{aligned}
$$

$$\boxed{Q_{\text{total}} = 344\,450\,\mathrm{J} \approx 344\,\mathrm{kJ}}$$

Notice: the latent heat (334 kJ) vastly exceeds the sensible heat (10.5 kJ) needed to warm the ice.

This is why ice is so effective at cooling drinks — most of the cooling comes from melting, not from warming the ice itself.

Example: Heating Ice to Steam

List the stages required to take 1.00 kg of ice at $-100\,°\mathrm{C}$ to steam at $300\,°\mathrm{C}$.

Stage	**Process**	**Formula**	**Specific value**
$-100\,°\mathrm{C} \to 0\,°\mathrm{C}$	Heat ice	$mc_{\text{ice}}\Delta T$	$c_{\text{ice}} = 2090\,\frac{\mathrm{J}}{\mathrm{kg}\,°\mathrm{C}}$
At $0\,°\mathrm{C}$	Melt	mL_f	$L_f = 334\,000\,\frac{\mathrm{J}}{\mathrm{kg}}$
$0\,°\mathrm{C} \to 100\,°\mathrm{C}$	Heat water	$mc_w\Delta T$	$c_w = 4186\,\frac{\mathrm{J}}{\mathrm{kg}\,°\mathrm{C}}$
At $100\,°\mathrm{C}$	Boil	mL_v	$L_v = 2\,256\,000\,\frac{\mathrm{J}}{\mathrm{kg}}$
$100\,°\mathrm{C} \to 300\,°\mathrm{C}$	Heat steam	$mc_s\Delta T$	$c_s = 2010\,\frac{\mathrm{J}}{\mathrm{kg}\,°\mathrm{C}}$

The two latent heat stages together account for the vast majority of the total heat required.

Check Your Understanding

How much more heat is needed to boil $1.00\,\mathrm{kg}$ of water (from $0\,^{\circ}\mathrm{C}$ to vapor) than to simply heat it from $0\,^{\circ}\mathrm{C}$ to $100\,^{\circ}\mathrm{C}$?

(Express the ratio.)

Come to lecture to see the answer.

Check Your Understanding

Burns from steam at 100 °C are typically much more severe than burns from liquid water at 100 °C.

Explain why, using the concept of latent heat.

Come to lecture to see the answer.

Part III

Heat Transfer Methods

Outline of Part III: Heat Transfer Methods

Overview of Heat Transfer

Conduction

Convection

Radiation

Outline of Part III: Heat Transfer Methods

Overview of Heat Transfer

Conduction

Convection

Radiation

Three Mechanisms of Heat Transfer

Conduction	Convection	Radiation
▶ Heat transfer through **direct physical contact**. ▶ Requires matter; no bulk movement of material. ▶ Examples: metal pot on stove, touching a cold railing.	▶ Heat transfer by **bulk movement of fluid** (liquid or gas). ▶ Requires matter; bulk material moves. ▶ Examples: boiling water, weather patterns, forced-air heating.	▶ Heat transfer via **electromagnetic waves** (infrared). ▶ Does *not* require matter. ▶ Examples: sunlight, campfire warmth, microwave oven.

In most real situations, all three mechanisms operate simultaneously.

Outline of Part III: Heat Transfer Methods

Overview of Heat Transfer

Conduction

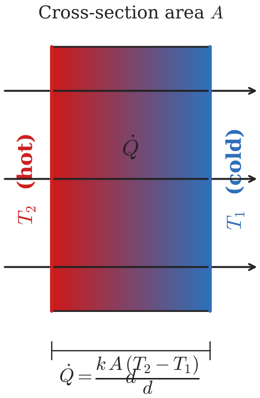
Convection

Radiation

Conduction

Mechanism

In conduction, heat moves through a material because high-energy (hot) molecules collide with and transfer energy to adjacent lower-energy (cool) molecules. No bulk motion of the material occurs.



- $\dot{Q}$: rate of heat transfer (W = J/s)
- k: thermal conductivity of material (J/(s·m·°C))
- A: cross-sectional area (m^2)
- $T_2 > T_1$: temperature difference (°C or K)
- d: thickness of material (m)

$\dot{Q} \propto A,\ \dot{Q} \propto \Delta T,\ \dot{Q} \propto 1/d$

Rate of Conductive Heat Transfer

$$\dot{Q} = \frac{Q}{t} = \frac{kA(T_2 - T_1)}{d}$$

Thermal Conductivity

Material	k **(J/s·m·°C)**
Silver	420
Copper	390
Aluminum	220
Steel/Iron	80
Concrete	1.3
Glass	0.84
Brick	0.6
Water	0.6
Wood	0.08–0.16
Air	0.023
Styrofoam	0.033
Wool/Insulation	0.04

The R-Factor

R-Factor (Thermal Resistance)

The R-factor is a measure of a material's thermal insulation:

$$R = \frac{d}{k}$$

Higher R = better insulator. Rewriting the conduction equation:

$$\dot{Q} = \frac{A(T_2 - T_1)}{R}$$

- In the US, insulation is often rated in "R-values" (in imperial units: $\mathrm{ft^2 \cdot {}^\circ F \cdot h/BTU}$).
- R-values of layers *add*: $R_{\text{total}} = R_1 + R_2 + \ldots$ (for layers in series).
- A double-pane window adds an air gap (k_{air} very small) to greatly increase R_{total}.

Example: Ice Cooler (ex. 14.5)

An ice cooler has walls with area $A = 0.950\,\mathrm{m}^2$, thickness $d = 2.50\,\mathrm{cm}$, and thermal conductivity $k = 0.0100\,\frac{\mathrm{J}}{\mathrm{s\,m\,{}^\circ C}}$. The inside is at $0\,^\circ\mathrm{C}$ and the outside is at $35.0\,^\circ\mathrm{C}$. How much ice melts in one day? ($L_f = 334\,000\,\frac{\mathrm{J}}{\mathrm{kg}}$)

$$t = 86\,400\,\mathrm{s} \quad (1 \text{ day})$$

$$Q = t \cdot \dot{Q} = t \cdot \frac{kA\Delta T}{d}$$

$$Q = \frac{(86\,400\,\mathrm{s})(0.0100)(0.950)(35.0)}{0.0250\,\mathrm{m}}$$

$$Q = 1.149 \times 10^6\ \mathrm{J}$$

$$m = \frac{Q}{L_f} = \frac{1.149 \times 10^6\ \mathrm{J}}{334\,000\,\frac{\mathrm{J}}{\mathrm{kg}}}$$

$$\boxed{m = 3.44\,\mathrm{kg}}$$

Over 3 kg of ice melts in a day, even with well-insulated walls.

Reducing d (thinner wall) or increasing A (bigger cooler) would melt more ice.

Check Your Understanding

If the thickness of an insulating wall is doubled while everything else stays constant, how does the rate of heat conduction change?

If the cross-sectional area is doubled, how does it change?

Come to lecture to see the answer.

Check Your Understanding

You can walk barefoot on hot sand at the beach but cannot stand on hot asphalt at a parking lot at the same temperature.

Similarly, you can briefly touch a wooden handle on a hot oven but not the metal rack.

Explain both observations using thermal conductivity.

Come to lecture to see the answer.

Outline of Part III: Heat Transfer Methods

Overview of Heat Transfer

Conduction

Convection

Radiation

Convection

Definition

Convection is heat transfer by **bulk motion of a fluid** (liquid or gas). Hot fluid rises (less dense), cool fluid sinks (more dense), creating circulation currents.

Natural Convection

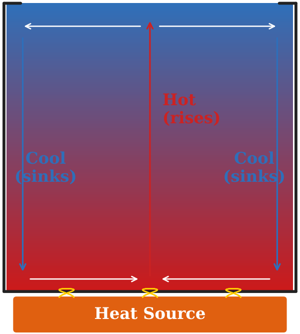
- Driven by buoyancy: hot fluid rises, cool fluid sinks.
- Examples: boiling water, atmospheric convection cells, ocean currents, lava lamps.
- Hot air rises in a room; heating vents are placed low.

Forced Convection

- Fluid is driven by external means (fan, pump).
- Examples: fan-forced ovens, car radiator with pump, air conditioning, blood circulation.
- More efficient than natural convection at transferring heat.

Convection in Biology and Nature

Natural Convection Cell



Examples in Nature

- **Wind**: air heated by land rises; cooler air flows in from sea (sea breeze).
- **Thunderstorms**: strong convective updrafts lift warm moist air.
- **Ocean gyres**: global heat transport via convection.
- **Earth's mantle**: slow convection drives plate tectonics.

Examples in Biology

- **Circulatory system**: blood (forced convection by heart) carries heat from core to extremities.
- **Fever**: increased blood flow to skin enhances heat loss.
- **Countercurrent exchange**: e.g., whale flippers keep warm blood near arteries to minimize heat loss.

Check Your Understanding

(a) Why is a fan useful in warm weather even though it does not cool the air?

(b) Why are heating vents placed near the *floor* and air-conditioning vents placed near the *ceiling*?

Come to lecture to see the answer.

Outline of Part III: Heat Transfer Methods

Overview of Heat Transfer

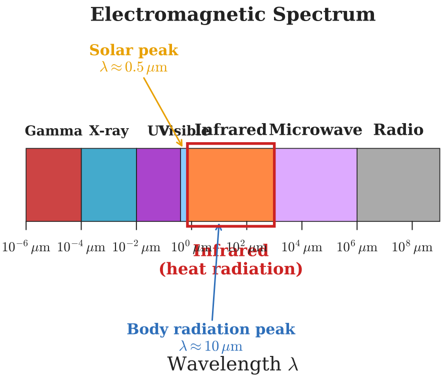
Conduction

Convection

Radiation

Radiation

Radiation is heat transfer via **electromagnetic waves** — primarily infrared radiation. No medium is required; radiation travels through vacuum.



- All objects with $T > 0\,$K emit radiation.
- Hotter objects emit more radiation (quartic T^4 dependence).
- Human body emits infrared ($T \approx 310\,$K).
- The Sun ($T \approx 5800\,$K) emits mostly visible light.

Everyday Examples

- Sunlight warming your face.
- Warmth felt from a campfire or incandescent bulb without touching.
- Infrared cameras detecting body heat.
- Earth cooling at night by radiating IR to space.

Stefan-Boltzmann Law

Net Rate of Radiative Heat Transfer

$$\dot{Q}_{\text{net}} = \sigma e A \left(T_2^4 - T_1^4 \right)$$

- $\dot{Q}_{\text{net}}$: net rate of heat emission (W); positive = net emission from object
- $\sigma = 5.67 \times 10^{-8}\,\mathrm{J/s/m^2/K^4}$: Stefan-Boltzmann constant
- e: emissivity of the object ($0 \leq e \leq 1$)
- A: surface area of the object ($\mathrm{m^2}$)
- T_2: temperature of the object (K); T_1: temperature of surroundings (K)

Temperature must be in kelvin. The T^4 dependence makes this extremely sensitive to temperature.

Emissivity e

Emissivity measures how efficiently an object radiates (and absorbs) thermal radiation relative to a perfect **blackbody** ($e = 1$).

- $e = 1$: perfect blackbody — absorbs and emits all radiation.
- $e = 0$: perfect reflector — absorbs and emits no radiation.
- Real objects: $0 < e < 1$

Material	**Emissivity** e
Human skin (all skin colors)	0.97–0.99
Dark rough surfaces (flat black paint)	0.90–0.98
Concrete/brick	0.85–0.95
Polished silver	0.02
Polished aluminum	0.03
White paint	0.85–0.95 (infrared, not visible!)

Example: Radiative Heat Loss (ex. 14.9)

Bob is unclothed in a dark room at 22.0 °C.
His skin temperature is 33.0 °C, emissivi ‘ty $e = 0.97$, and surface area $A = 1.50\,\mathrm{m}^2$.
At what rate does Bob lose energy by radiation?

$$T_2 = 33.0 + 273.15 = 306.15\,\mathrm{K}$$
$$T_1 = 22.0 + 273.15 = 295.15\,\mathrm{K}$$
$$\begin{aligned}\dot{Q}_{\mathrm{net}} &= \sigma e A(T_2^4 - T_1^4) \\ &= (5.67 \times 10^{-8})(0.97)(1.50) \\ &\quad \times \left(306.15^4 - 295.15^4\right) \\ &= 8.25 \times 10^{-8} \times 1.196 \times 10^9\end{aligned}$$

$$\boxed{\dot{Q}_{\mathrm{net}} \approx 98.8\,\mathrm{W} \approx 100\,\mathrm{W}}$$

Bob radiates roughly **100 W** — comparable to a 100-W incandescent bulb!

This is why a crowded room feels warm: many bodies radiating ∼100 W each quickly heat the space.

Check Your Understanding

A person's net radiation rate is *zero* when they are in thermal equilibrium with their surroundings.

At what body temperature does this occur if $T_{\text{surroundings}} = 37\,^{\circ}\text{C}$?

Does this match normal body temperature? What does this imply?

Come to lecture to see the answer.

Check Your Understanding

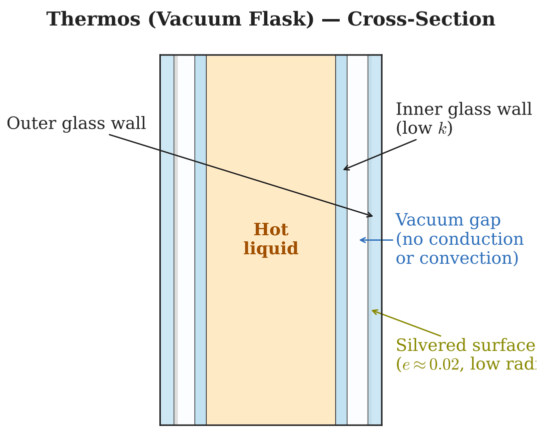
In bright sunlight, should you wear dark or light clothing to stay *cool*?

What about at night in a desert when you want to stay *warm*?

(Hint: emissivity applies to both absorption and emission of radiation.)

Come to lecture to see the answer.

Thermos Bottles: Minimizing All Three Heat Transfer Mechanisms



The thermos bottle is a near-perfect insulator precisely because it addresses *all three* heat transfer mechanisms simultaneously.

The vacuum is the most important feature: eliminating conduction and convection through air accounts for most of the insulation.

Design Features

- **Double glass walls**: minimal conduction through glass (k small).
- **Vacuum gap**: eliminates conduction and convection through air.
- **Silvered inner surface**: low emissivity ($e \approx 0.02$) minimizes radiation.

Chapter 14 Summary

Topic	Equation	Condition
Heat & temperature change	$Q = mc\Delta T$	Single phase
Heat & phase change	$Q = mL$	At phase boundary
Calorimetry	$\Sigma Q = 0$	Isolated system
Conduction rate	$\dot{Q} = \dfrac{kA(T_2 - T_1)}{d}$	Steady state
R-factor	$R = d/k$	Higher R = better insulator
Radiation rate	$\dot{Q}_{\text{net}} = \sigma e A(T_2^4 - T_1^4)$	T in kelvin
$\sigma = 5.67 \times 10^{-8}\ \mathrm{J/s/m^2/K^4}$,	$1\,\mathrm{k} = 4186\,\mathrm{J}$	